

Experiment-12:

Implement Iris Flower Classification using KNN .

Code:

```
# =====  
# Iris Flower Classification using K-Nearest Neighbors (KNN)  
# =====  
  
# Step 1: Upload Dataset  
from google.colab import files  
uploaded = files.upload()  
  
# Step 2: Import Libraries  
import pandas as pd  
import numpy as np  
import matplotlib.pyplot as plt  
import seaborn as sns  
  
from sklearn.model_selection import train_test_split  
from sklearn.preprocessing import LabelEncoder  
from sklearn.neighbors import KNeighborsClassifier  
from sklearn.metrics import accuracy_score, confusion_matrix, classification_report  
  
# Step 3: Load Dataset  
df = pd.read_csv("Iris.csv")  
print("First 5 Rows:")  
print(df.head())  
print("\nDataset Shape:", df.shape)  
  
# Step 4: Remove ID Column (if available)  
if 'Id' in df.columns:  
    df.drop('Id', axis=1, inplace=True)  
  
# Step 5: Encode Species Labels  
encoder = LabelEncoder()  
df['Species'] = encoder.fit_transform(df['Species'])
```

```

print("\nEncoded Dataset:")

print(df.head())

# Step 6: Split Dataset
X = df.iloc[:, :-1]
y = df.iloc[:, -1]
X_train, X_test, y_train, y_test = train_test_split(
    X, y,
    test_size=0.2,
    random_state=42
)

# Step 7: Create KNN Model
knn = KNeighborsClassifier(n_neighbors=3)

# Train Model
knn.fit(X_train, y_train)

# Step 8: Prediction
y_pred = knn.predict(X_test)

# Step 9: Accuracy
accuracy = accuracy_score(y_test, y_pred)
print("\nAccuracy: {:.2f}%".format(accuracy * 100))

# Step 10: Confusion Matrix
cm = confusion_matrix(y_test, y_pred)
plt.figure(figsize=(6,5))
sns.heatmap(cm,
    annot=True,
    cmap="Blues",
    fmt='d',
    xticklabels=encoder.classes_,
    yticklabels=encoder.classes_)
plt.xlabel("Predicted")
plt.ylabel("Actual")

```

```

plt.title("Confusion Matrix")

plt.show()

# Step 11: Classification Report
print("\nClassification Report:\n")

print(classification_report(
    y_test,
    y_pred,
    target_names=encoder.classes_
))

# Step 12: Predict New Flower
sample = [[5.1, 3.5, 1.4, 0.2]]
prediction = knn.predict(sample)
print("Predicted Flower Species:",
      encoder.inverse_transform(prediction)[0])

```

Output:

...

Choose Files
Iris.csv

Iris.csv(text/csv) - 5107 bytes, last modified: 8/5/2026 - 100% done
Saving Iris.csv to Iris.csv
First 5 Rows:

	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm	Species
0	1	5.1	3.5	1.4	0.2	Iris-setosa
1	2	4.9	3.0	1.4	0.2	Iris-setosa
2	3	4.7	3.2	1.3	0.2	Iris-setosa
3	4	4.6	3.1	1.5	0.2	Iris-setosa
4	5	5.0	3.6	1.4	0.2	Iris-setosa

Dataset Shape: (150, 6)

Encoded Dataset:

	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm	Species
0	5.1	3.5	1.4	0.2	0
1	4.9	3.0	1.4	0.2	0
2	4.7	3.2	1.3	0.2	0
3	4.6	3.1	1.5	0.2	0
4	5.0	3.6	1.4	0.2	0

Accuracy: 100.00%

Classification Report:

	precision	recall	f1-score	support
Iris-setosa	1.00	1.00	1.00	10
Iris-versicolor	1.00	1.00	1.00	9
Iris-virginica	1.00	1.00	1.00	11
accuracy			1.00	30
macro avg	1.00	1.00	1.00	30
weighted avg	1.00	1.00	1.00	30

Predicted Flower Species: Iris-setosa